

Trends in Nationalization of State and Local Elections: Evidence from Precinct Data

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Abstract

According to recent work, the nationalization of state and local politics varies by the level of office, content of the contest, and presence of local media. Was politics always this nationalized? We use precinct-level election results from 13 states over the past 25 years to study the relationship between national, state, and local contests. Relying on evidence from 23,017 contests, including many nonpartisan races and ballot measures, we show that partisanship in down ballot contests has been steadily rising since 2000. Although national partisanship explained 95 percent of the within-contest variation in voting for statewide partisan contests in 2024, it only explained 84 percent in 2000. In statewide nonpartisan contests the difference is similar: 28 percent in 2024 versus 18 percent in 2000. In local partisan and nonpartisan contests the percentage has remained stable since 2000. We emphasize that although nationalization in down ballot contests is slowly trending upwards, nationalization remains relatively low and the largest differences are still between nonpartisan and partisan contests.

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Introduction

The nationalization of politics at the state and local level is an animating theme of recent research in political science. By nationalization we mean that among voters who select the party P candidate at the top of the ballot are increasingly likely to also select the party P candidate in down ballot contests. Although scholars continue to debate *why* nationalization occurs, it is clearly present in a wide array of electoral contests up and down the ballot. For certain time periods in American history nationalization has even declined, but since at least 1980 voters are increasingly likely to select candidates from the same party at the state and national level (Hopkins 2018). What about local contests? Most contemporary research focuses either on single offices or has short time horizons, leaving us with many questions about the historical trend of partisan divisions in local politics. In this article, we provide new evidence of a historical narrowing between local and national vote choice, although the difference remains larger than statewide contests.

Until recently, local politics was theorized to be largely different than national politics. The distinctiveness of the work subnational governments do – often nonpartisan and nonideological in nature with a focus on economic growth and development – would cast doubt on the expectation that vote choice in local elections would fall along any sort of partisan dimension (Peterson 1981; Oliver, Ha, and Callen 2012; Anzia 2021). Interest group cleavages at the local level often differ from national cleavages, perhaps offering a cross-cutting cue to voters (Anzia 2022; Anzia and Trounstein 2024; DiSalvo 2023). Finally, voters may have direct experience with local government services or know candidates personally, unlike national politics and politicians. When comparing local politics to state politics, we expect these differences to persist.

One comprehensive study uses ballot data from the 2020 election to compare partisan divisions at all levels of government (Conevska et al. 2025). At the state level, their results confirm the stark partisan divisions presented in Hopkins (2018). At the local level, they find stark divisions in partisan contests, but in nonpartisan contests (including ballot

questions) divisions shrink dramatically. Other research on local politics tends to focus on single offices and has, perhaps as a consequence, found mixed results on the presence of partisanship.¹

Our article seeks to extend this question back in time. Have divisions at the local level always existed in partisan contests, but to a lesser degree in nonpartisan contests or ballot questions? **Overview of results**

Subpoint details. For example: We also extend the analysis to heterogeneous treatment effects. For example, do local partisan divisions increase more in areas with more local media? Do they increase more in areas with more competitive local elections? Big urban areas vs small rural towns? Type of issue?

Although our preference would be to rely on ballot level data, akin to Conevska et al. (2025), such data has only recently become available to researchers (Kuriwaki et al. 2024). Instead, we rely on aggregate precinct level results from elections conducted since the turn of the century in 13 states. Precinct level returns are the most granular results available to researchers in the United States since around the turn of the century. Our aggregate results cover 19 election years and 23,017 unique contests. We validate our use of aggregate data by comparing Goodman style regressions using precinct data to the absolute truth found in ballot data.² We find that when more than ten precincts cast ballots in a given contest and when the results of that contest are not heavily skewed towards a given candidate, our regression coefficients correlate highly with the truth measured using the ballot data. To our knowledge, we are the first to empirically validate the use of aggregate data and set reasonable bounds on where it is likely to be sufficient for making inferences. We hope our guidance will help extend the study of local elections

1. For example, in local executive offices Schaffner, Streb, and Wright (2001), Taylor and Schreckhise (2003), and Alvarez, Hall, and Levin (2018), state legislatures (Schaffner, Streb, and Wright 2001; Ansolabehere et al. 2006; Garlick 2015), judicial offices (Squire 1988; Klein and Baum 2001; Rock and Baum 2010; Burnett and Tiede 2015; Bonneau and Cann 2013; Lim and Snyder 2015; Kritzer 2021; Weinschenk et al. 2021), and school boards (Weinschenk 2022).

2. Although here we focus on aggregate precinct-level data, we also discuss county-level data in subsection C. In short, county-level data is appropriate to study individual level data only in very specific situations.

back in time before ballot data become available, or in jurisdictions where ballot data is still unavailable.

Data

Not all state governments collect and publish election returns at the precinct level for local contests. However, eight states – Arizona, Maryland, Massachusetts, Nevada, Oregon, Rhode Island, Texas, and South Carolina – do publish such data going back in time. Additionally, we collect local data from seven counties – Santa Clara, California; Marion, Indiana; Cuyahoga, Ohio; Franklin, Ohio; Hamilton, Ohio; King, Washington; and Pierce, Washington. For some places, we were able to collect precinct level election returns for a similar time period, but only for state level contests. These data come from Alaska, California, Colorado, Idaho, and Michigan.

We compiled and organized the precinct level results. Essential information about each contest and candidate is often missing, and must be obtained and manually incorporated from other sources. This includes information about district magnitude, whether races were partisan or nonpartisan, the party labels of the candidates, and the content and topics of the ballot measures.

The result is a dataset of thirteen entire states and seven counties, covering all general elections held in even numbered years between, roughly, the turn of the century and 2024. The exact range of data available for each state or county is documented in Figure A.1. Our data includes precinct level election returns for all state and local partisan offices, all state and local nonpartisan offices, and all state and local ballot measures. This data is necessarily a convenience sample, as it is limited to the states and counties that publish precinct level returns for local contests in a machine readable format going back in time. Nevertheless, the places in our sample are geographically and politically diverse, and vary in the types of local contests on their ballot.

Validating the Use of Precinct Level Returns

We are primarily interested in the relationship between an individual voter’s decision in national or state races a top of the ticket contest and the voter’s vote in state or local contests further down the ballot.³ Clearly, the ideal data for this purpose would be at the individual level. Unfortunately, these data are only available in very limited circumstances, and have only begun to become widely available after 2020. Instead, we turn to aggregated election returns at the precinct level. The use of this type of data has a long historical tradition. However, as has been known for nearly as long, using aggregate level results to infer behavior about individual decisions suffers from the ecological inference problem (Goodman 1953; King 1997). We do not offer a new solution to this problem but we validate the accuracy of our estimates from aggregate data using ballot level data where it is available.

To fix ideas, consider the following model. Let k index contests, let i index individuals, and let j index precincts. Let $V_{kij} = 1$ if voter i (who votes in precinct j) chooses the Democratic alternative in contest k and $V_{kij} = 0$ if i chooses the Republican alternative. Let D_{ij} be a measure of voter i ’s partisanship based on national or top-of-the-ticket offices.⁴ Suppose vote choices are determined by:

$$V_{kij} = \alpha_k^I + \beta_k^I D_{ij} + \epsilon_{kij} \quad (1)$$

So, α_k^I captures valence characteristics contest k , and β_k^I captures partisan characteristics. If D_{ij} is dichotomous, with $D_{ij} = 1$ indicating Democratic voters and $D_{ij} = 0$ indicating

3. We also extend this question to combinations of top of the ticket contests and/or bottom of the ticket contest combinations.

4. We consider the top-of-the-ticket office in each year to be drawn from the following choices, in descending order of preference: U.S. President, U.S. Senate, State Governor. In Appendix C we validate this choice and show that our main results are robust to alternative specifications of this choice.

Republican voters, then

$$\alpha_k^I = \text{Prob}(V_{kij}=1 \mid D_{ij}=0) \quad (2)$$

$$\beta_k^I = \text{Prob}(V_{kij}=1 \mid D_{ij}=1) - \text{Prob}(V_{kij}=1 \mid D_{ij}=0) \quad (3)$$

For our purposes, the main parameter of interest is β_k^I , which measures the difference between Democrats and Republicans in voting in contest k .

Next, for each precinct j , average over all of the N_j voters in j to create the variables $V_{kj} = (1/N_j) \sum_{i=1}^{N_j} V_{kij}$ and $D_j = (1/N_j) \sum_{i=1}^{N_j} D_{ij}$. Consider the model:

$$V_{kj} = \alpha_k^P + \beta_k^P D_j + \epsilon_{kj} \quad (4)$$

If equation 1 is the true model and the aggregation of voters into precincts does not depend on the ϵ_{kij} 's, then $\beta_k^I = \beta_k^P$ (and $\alpha_k^I = \alpha_k^P$).

check

However, as is well known, if equation 1 is not the true model, or if voters are aggregated into precincts systematically depending on the ϵ_{kij} 's, then β_k^I and β_k^P can be very different (as can α_k^I and α_k^P).

For example, suppose that in addition to partisanship, some voter are affected by whether a candidate lives near them – i.e., some amount “friends and neighbors” or “hometown” effect (**key_1949**; **rice_1987**). Let L_{kij} be 1 if voter i lives closer to the Democratic candidate in contest k , and 0 if they live closer to the Republican candidate in contest k . Suppose $\epsilon_{kij} = \gamma L_{kij} + \mu_{kij}$. If L_{kij} is uncorrelated with D_{ij} then least squares estimates of equation 1 will produce $\text{plim } \hat{\beta}_k^I = \beta_k^I$. Aggregating to the precinct level, $L_{kj} = (1/N_j) \sum_{i=1}^{N_j} L_{kij}$, gives the fraction of voters who live closer to Democratic candidate in contest k . Given geographic sorting and career paths in which candidates for higher office often have served in lower level offices, it is likely that L_{kj} and D_j are positively correlated. For example, this will occur if Democratic candidates come from

areas with more Democratic voters and Republican candidates come from areas with more Republican voters. In this case, the least squares estimates of equation 4 will be biased upward; and therefore $\text{plim } \hat{\beta}_k^P > \text{plim } \hat{\beta}_k^I$.

For example, suppose β_k^I varies across voters – in particular, that it is higher for liberal Democrats compared to moderate Democrats and lower for conservative Republicans compared to moderate Republicans. Suppose further that there is geographic sorting, so that when D_j is higher, the ratio of liberal to moderate Democrats is higher and/or the ratio of moderate to conservative Republicans is higher. Then the estimated value of β_k^P will tend to be larger than the estimated value of β_k^I (which is itself a weighted average of the high and low β_k^I). XXX DO WE NEED TO ADD MORE HERE? OTHER CASES? XXX

We use ballot level data, commonly called cast vote records (CVR), from 2020 to uncover the conditions under which the point estimates of the β_k^P 's from the precinct level regression are similar to the point estimates of the β_k^I 's from ballot-level regressions. That is, we estimate Equation 1 using the CVRs. We then aggregate the CVR records by averaging V and D over all voters in each precinct. We then estimate Equation 4 using the precinct-level averages. We then compare the estimated β_k 's. We do this separately for six types of contests: state partisan races, local partisan races, state nonpartisan races, local nonpartisan races, state ballot measures, and local ballot measures.⁵

Figure 1 shows scatterplots of the precinct-level slope estimates (y-axis) versus the individual-level slope estimates (x-axis). In five of the six cases, the correlations are quite high, above 0.76. The correlation coefficients for state partisan races is noticeably lower, in part because there is little variation in $\hat{\beta}_k^I$. For nonpartisan races and ballot measures, the correlation coefficients are higher at the state level compared to the local level. This in part because the number of precincts tends to be much larger at the state level, and the standard deviation of D_j is also larger on average.

5. For partisan races, we only use cases with one Democratic candidate and one Republican candidate, and where no third-party or independent candidate received more than 20% of the total votes cast or placed first or second. For nonpartisan races, we keep the top two vote receivers. For ballot measures, we only use the yes or no choice.

Note that in Figure 1 we only include contests with five or more precincts and for which the standard deviation of D_j across precincts is at least 0.10. Appendix Figure B.1 helps justify these choices. The figure is a heatmap of correlation coefficients. The x-axis shows the threshold minimum number of precincts, and the y-axis show the threshold minimum standard deviation of D_j . Except for state partisan races, once the minimum standard deviation of D_j is 0.1 or higher, and the number of precincts is 5 or more, the correlations are virtually all above 0.7 and the vast majority are above 0.8.

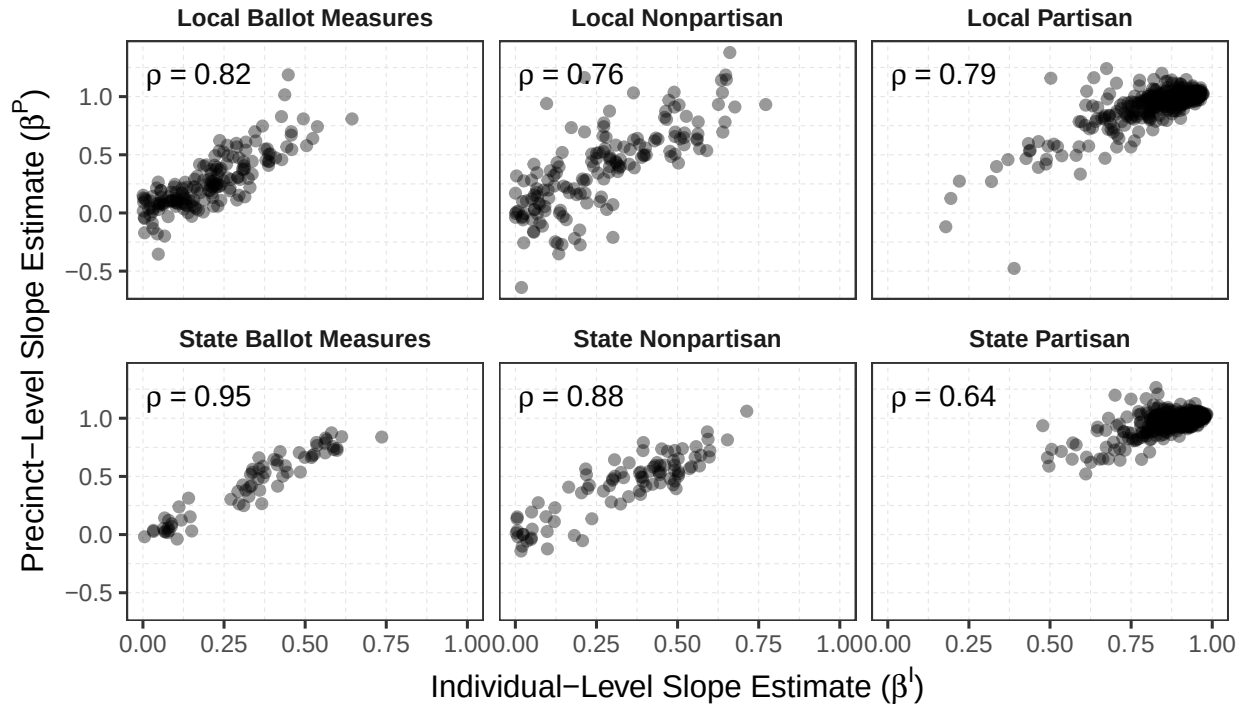


Figure 1: β^P Versus β^I for Each Type of Contest
(min. standard deviation of $D = .10$, min. number of precincts = 5)

For all types of contests, $\hat{\beta}_k^P$ tends to be larger than $\hat{\beta}_k^I$ for higher values of $\hat{\beta}_k^I$. For example, for local ballot measures, when $\hat{\beta}_k^I$ is greater than 0.25 the average level of $\hat{\beta}_k^I$ is 0.36 where as the average of $\hat{\beta}_k^P$ is 0.49. For local nonpartisan races, the corresponding averages are 0.42 and 0.62, respectively. For state ballot measures, the corresponding averages are 0.44 and 0.58 and for state nonpartisan races, 0.45 and 0.57. These patterns are consistent with geographic sorting of some type – e.g., based on partisanship or

ideology. For some contests, they might also reflect voting on the basis of candidate attributes, such as “friends and neighbors” voting.

Using exit poll data, we are able to check whether or not the difference between $\hat{\beta}_k^P$ and $\hat{\beta}_k^I$ has been noticeably trending over time. Appendix Figure XX shows that for state ballot measures, there is no obvious trend. We do not have individual-level data about vote choice for local ballot measures, local nonpartisan offices, or state nonpartisan offices (these are not included in exit polls). However, since the distributions of $\hat{\beta}^I$ for these three types of contests are roughly similar to the distribution for state ballot measures in 2020, there is no obvious reason to expect that the differences, $\hat{\beta}^P - \hat{\beta}^I$, would exhibit strong trends.

Overall, precinct level data can be used to estimate the strength of national partisanship on voting with a relatively high degree of confidence provided: (i) the number of precincts is large enough, and (ii) the variation in D_j across the precincts is large enough. These are intuitive conditions. Based on the above patterns, we restrict all analyses below to contests with at least five precincts and for which the standard deviation of D_j is at least 0.10.

Nationalization in Local and State Contests Over Time

Figure 2 presents our main estimates of nationalization in local and state contests over time. Although our data comprises of some estimates from the 1990s, the vast majority of our data lies in the range 2000-2024 and we limit our figures to this period. In Appendix D we separately reproduce the main figure for each jurisdiction with both state and local data, where we show the full range of each place. On the y-axis is the estimate of β^P from the precinct level regression for each contest. Each point is a single contest. The red dashed line indicates the theoretical maximum value of β^P , which is 1. The blue and green lines are a smoothed trend line across all contests. When computing the blue line,

we weight each contest point by the number of voters in the contest, so that contests with more precincts have more influence on the trend line. The green line is a smoothed trend line across all contests without weighting. In our interpretation, the blue line helps evaluate the “average voter,” while the green line helps evaluate the “average contest.”

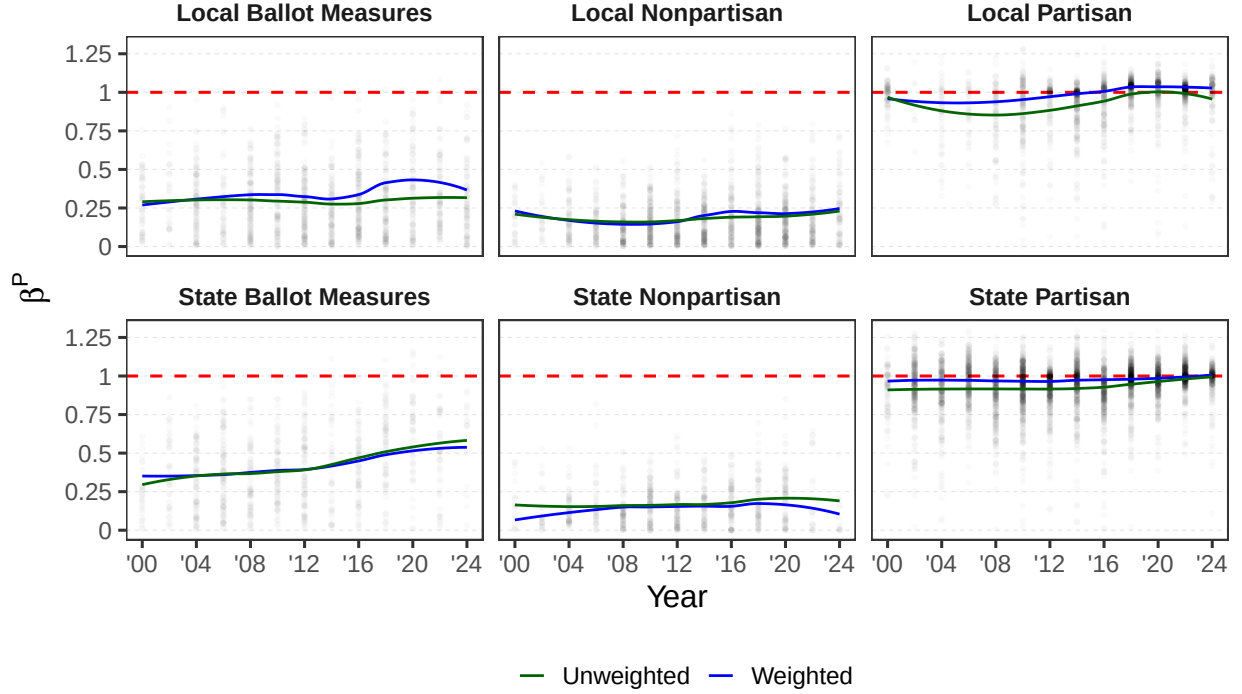


Figure 2: **Main Results**

For local ballot measures and nonpartisan races, the average estimated β^P s are low. The estimates for local ballot measures do not appear to be trending over time. For local nonpartisan races, there appears to be a slight upward trend from 2000 to 2008, after which there is little change. For local partisan races the estimates are on average high – near 1 – and with no clear upward trend. Regarding the levels in 2020, the patterns are similar to those in Conevska et al. (2025).

For state ballot measures, the average estimated β^P is again relatively low. Here, however, compared to local ballot measures, there is clear upward trend. One caveat is that the variation across years might be influenced by the agenda – e.g., the average estimated β^P is likely to be higher if more of the measures on the ballot in a given year

involve polarizing issues, such as abortion or regulations connected to climate change. For state nonpartisan races the average estimated β^P is again relatively low but appears to be trending slowly upwards. Around 2016 the weighted curve begins to trend upwards at a quicker rate than the unweighted curve. This could perhaps indicate that in places with more voters, the effect of nationalization on state nonpartisan races is more pronounced. For state partisan races, the average estimates are again near 1 with no clear upward trend.

Variation Across Offices

Although we do not find clear trends over time for offices other than ballot measures, this could be masking important heterogeneity across the type of office. Thus, we categorize each contest (excluding ballot measures) into one of six categories: education, executive, judicial, law enforcement, legislative, or school board. We then separate these office types into partisan and nonpartisan contests, and plot the estimated β^P over time for each category in Figure 3.

Ballot Measures

While we find little evidence of an overall time trend in β^P for local ballot measures and only a modest upward trend for state measures, there is some heterogeneity across issue areas. Figure 4 plots average estimated β^P over time by ballot measure content. Several issue areas exhibit clear upward trends, including community and economic development, drugs (e.g., cannabis and alcohol sales), governance (e.g., charter amendments), social issues (e.g., abortion and same-sex marriage), and spending (e.g., bonds, tax changes, and budgets). Measures related to development, social issues, and spending consistently display higher average β^P values than the overall mean across all measures (black dashed line), indicating greater partisan polarization. By contrast, governance measures—which

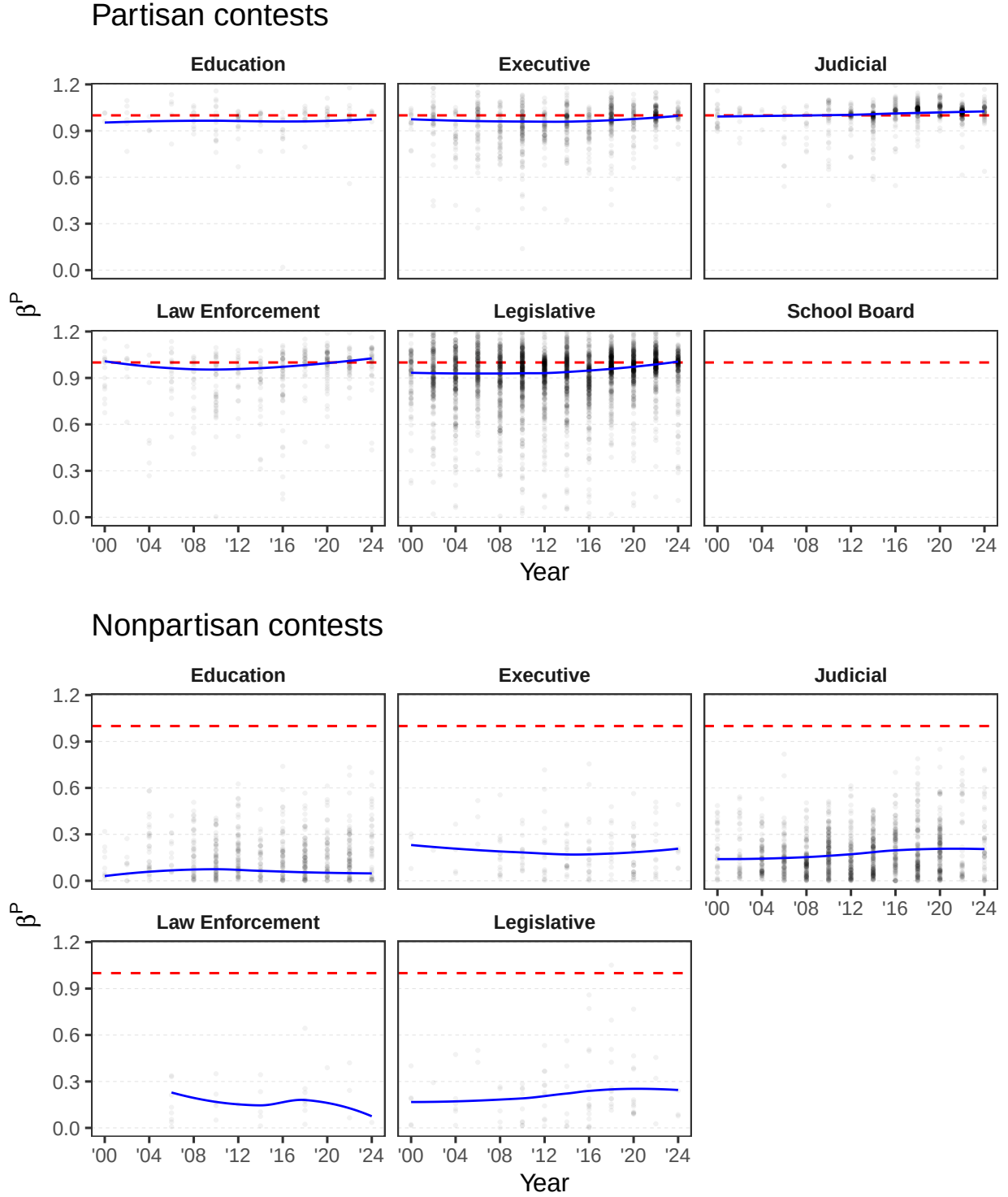


Figure 3: **Office types.** Plots the estimated β^P over time for partisan (top) and nonpartisan (bottom) contests, divided into office types. The blue line is a loess fit weighted by the number of votes in each contest. Otherwise, the plots are constructed in the same manner as Figure 2.

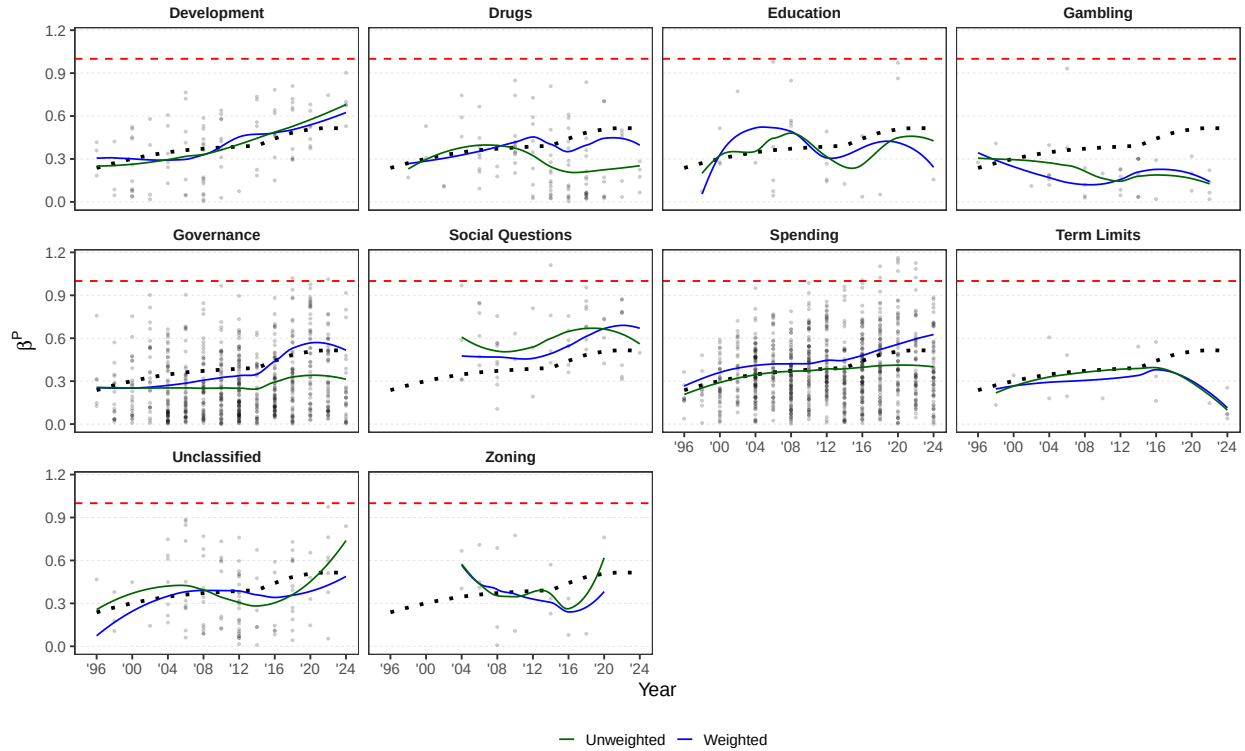


Figure 4: **Ballot Measures by Issue Area.** Plots the estimated β^P over time for ballot measures, divided into issue areas. Blue lines are loess smoothed trend lines for the specified issue area; black dashed lines are trend lines across all measures and can be used as a relative reference point for an individual category. The red dashed line indicates the theoretical maximum value of β^P .

tend to be more technical—exhibit lower average β^P throughout the period. For the remaining categories—education, gambling, term limits, and zoning—the limited number of observations precludes strong conclusions.

Where we do have a notable amount of observations, however, is in spending ballot measures. Figure 5 disaggregates these measures by the types of projects they fund. Although small sample sizes within some subcategories limit the sort of claims we can make, no single spending category appears to drive the modest upward trend in average β^P for spending measures overall. Instead, the key distinction lies in which types of spending are relatively more or less partisan. Measures funding core local government functions such as infrastructure, public safety, and libraries are consistently less partisan, or roughly in line with the overall spending average. In contrast, spending measures

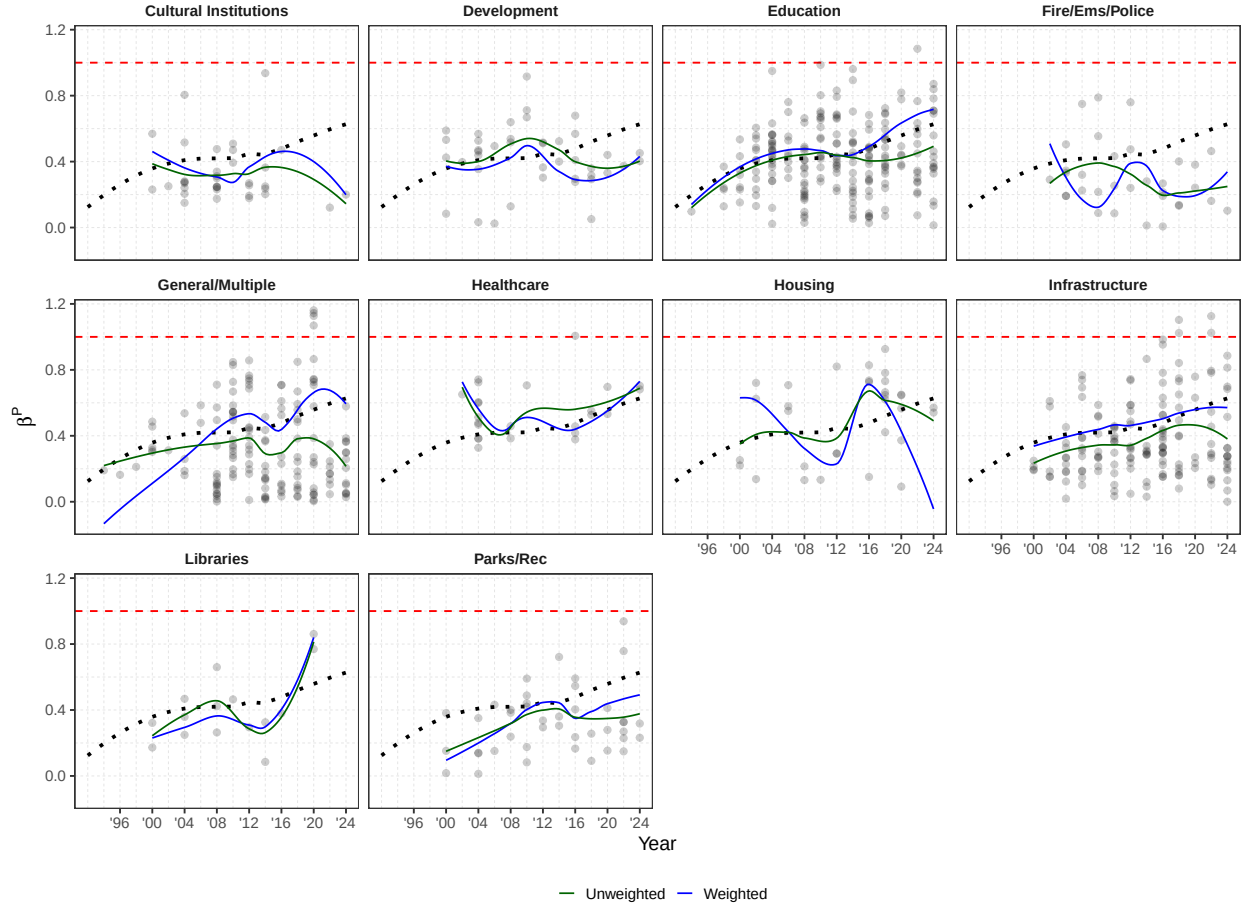


Figure 5: **Spending Measures, by Target.** Plots the estimated β^P over time only for ballot measures about spending, divided into the target of each spending measure. The black line plots the overall trend of all ballot measures about spending, for reference. Blue and red lines are constructed similarly to Figure 4.

related to community development (e.g., parks and recreation), education, and housing are markedly more partisan across years. In neither case does the gap between types of spending measures appear to change over time.

Conclusion

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Appendix for "Trends in Nationalization of State and Local Elections"

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A Data

We collect election returns at the precinct-level from a variety of sources. Where available, we always rely on official state sources publicly available online. In some cases, we contact state officials directly to obtain precinct-level returns. The range of our data is documented in Figure A.1. Most of the ballot data, or cast vote records, was collected as part of a broader project and are documented in Kuriwaki et al. (2024). We also combine this data with cast vote records from multiple years of South Carolina elections, released as part of Kuriwaki (2025).

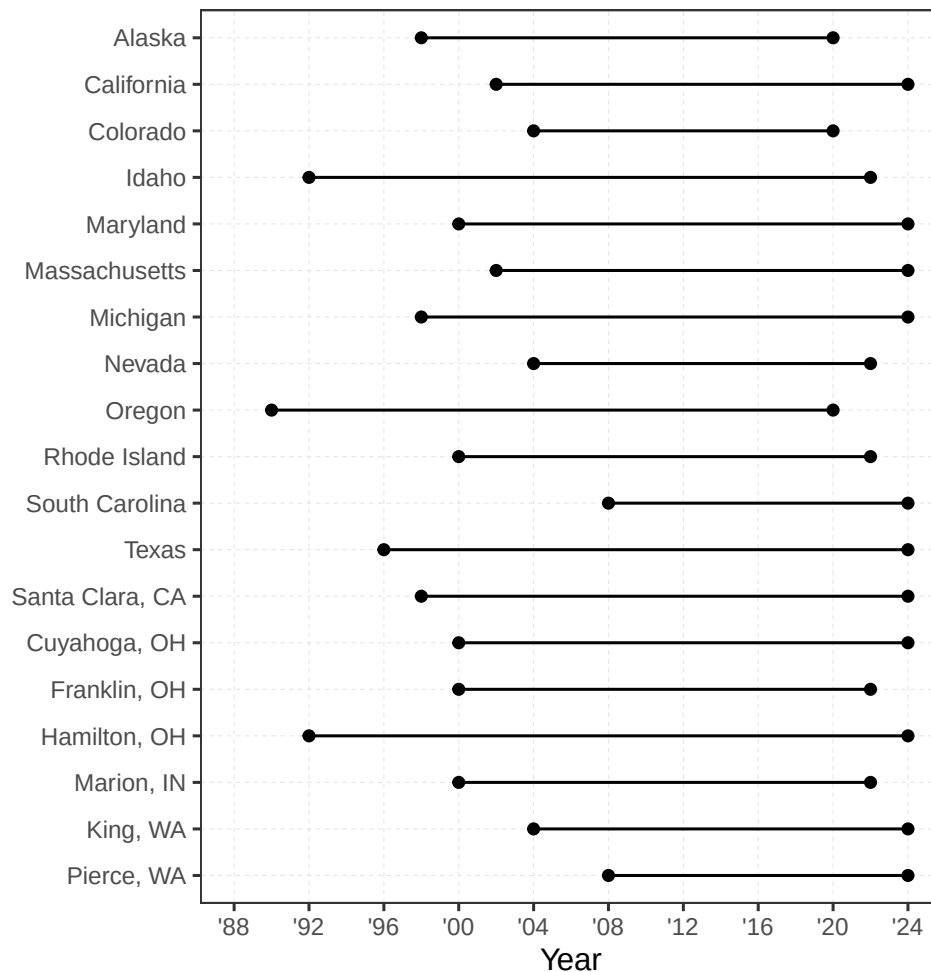


Figure A.1: **Range of Precinct Data.** For each state and county in our sample, we document the range of years for which we have precinct level data. [TODO: Note which states we have state AND local data, perhaps using a color?]

B Additional Statistical Results

A Correlations between β^p and β^l

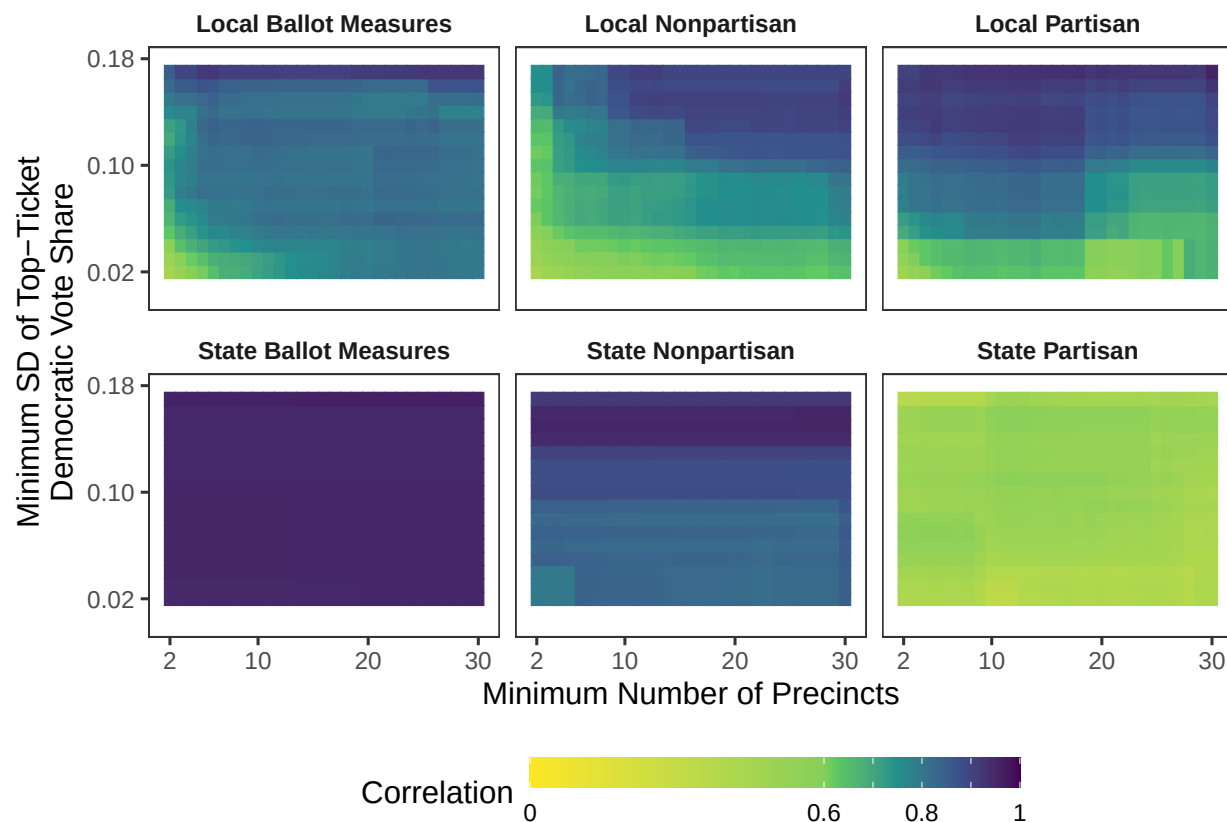


Figure B.1: Correlation Between Precinct Level and Ballot Level Estimates by Minimum Number of Precincts and SD of Vote Share

B T-test of Over Time Trends

C County Level Data

D Nationalization and the Size of the Contest

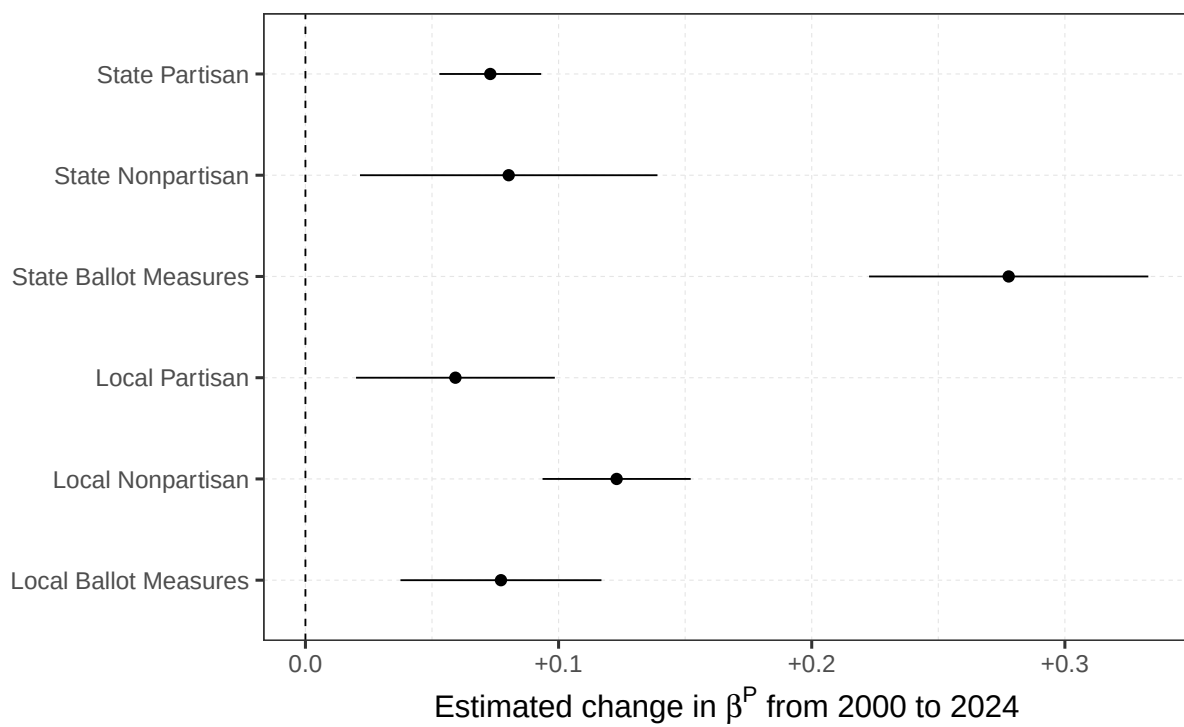


Figure B.2: **T-test of Over Time Trends in β^P** . For each contest type, we test whether the average β^P in 2000 is different from the average β^P in 2024. The figure displays the p-value for a two-sided t-test of the difference in means, as well as the difference in means and the 95% confidence interval for that difference.

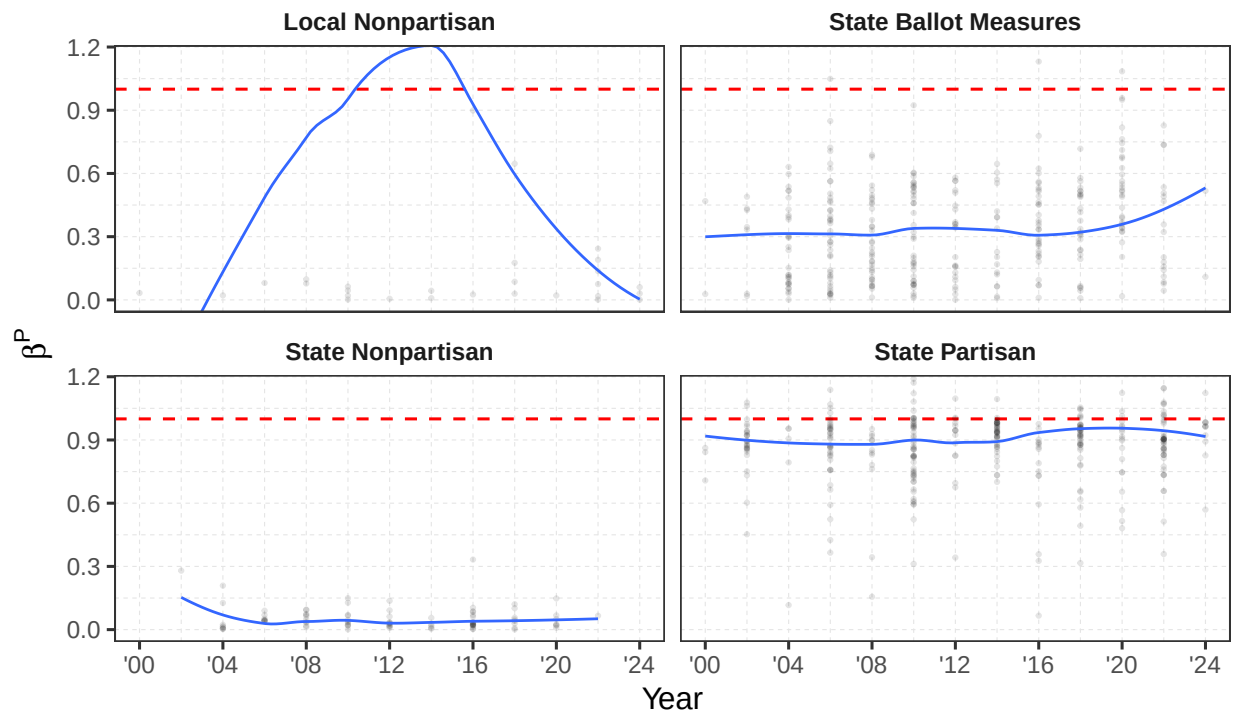


Figure B.3: **Figure 2 Using County-Level Data.** Figure constructed in same way as Figure 2, but using county-level data instead of precinct-level data. Note we can only use statewide contests because of the lack of variation for sub-state contests when using county-level data.

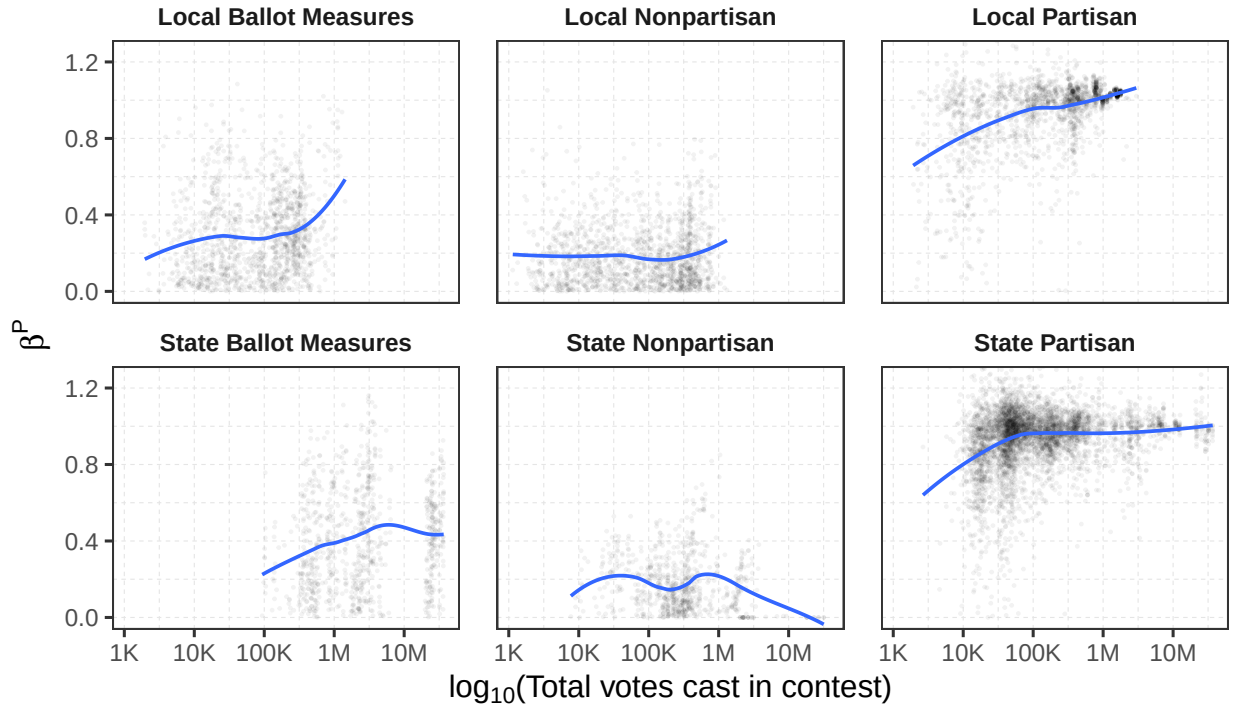


Figure B.4: **Nationalization of State and Local Elections by the Size of the Contest.** We report β^P for each contest type, by the size of the contest, as measured by the number of votes cast for each contest. Blue lines are smoothed trend lines fit to the points. Note that the x-axis is reported on the log scale.

C Robustness to the Choice of Top-of-the-Ticket Office

Since we are making a claim about nationalization, it makes most sense to use national elections as the reference category to compare against. We prefer to use vote totals for president where available. In off-cycle years, we prefer to first use totals for the U.S. Senate, then if those are unavailable, votes for the state's governor. The specific choices we make in each jurisdiction for non-presidential years can be found in Table C.1.

Table C.1: **Top-Ticket Office In Each State and Off-Cycle Election.** S = U.S. Senate; G = State Governor. The table reports the highest ticket office elected in each state in our data, for every off-cycle election where we have data. In presidential years, the top-ticket office for all states is the U.S. President.

	'90	'94	'98	'02	'06	'10	'14	'18	'22
AK	-	-	S	S	G	S	S	G	-
CA	-	-	S	G	S	S	G	G	S
CO	-	-	-	-	G	S	S	G	-
ID	-	G	S	S	G	S	S	G	G
IN	-	-	-	-	-	S	-	S	S
MA	-	-	-	G	S	G	S	S	G
MD	-	-	-	-	S	S	G	S	S
MI	-	-	G	S	S	G	S	S	G
NV	-	-	-	-	S	S	G	S	S
OH	-	S	S	G	S	S	G	S	S
OR	S	G	S	S	G	S	S	G	-
RI	-	-	-	S	S	G	S	S	G
SC	-	-	-	-	-	S	S	G	S
TX	-	-	G	S	S	G	S	S	G
WA	-	-	-	-	S	S	-	S	S

We validate this choice in several ways. First, the correlation in the vote totals between these offices are all high. Where available, we compute the correlation between president-senate, president-governor, and senate-governor vote totals at the precinct level. The median correlations are 0.981 (president-senate), 0.976 (president-governor), and 0.971 (senate-governor). We caution against interpreting these results too strongly, but taken together they suggest that our choice of top-of-the-ticket office is not driving our results.

Second, we re-estimate our main results using limited combinations of top-of-the-ticket offices. Specifically, we re-estimate our main results using only presidential years, and then again using only presidential years plus off-cycle years with Senate contests. The results are substantively similar to those in the main text, suggesting that our choice of top-of-the-ticket office is not driving our results. These results are presented in Figures C.1 and C.2.

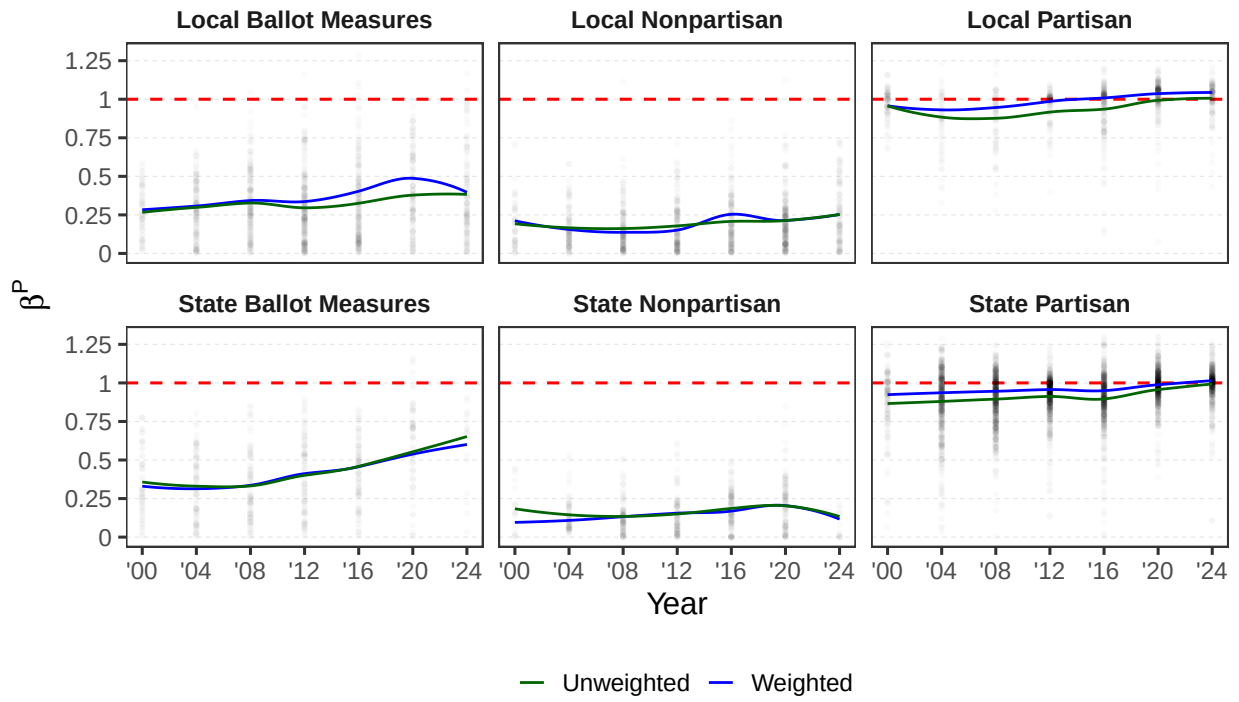


Figure C.1: Nationalization of State and Local Elections Using Only Presidential Years

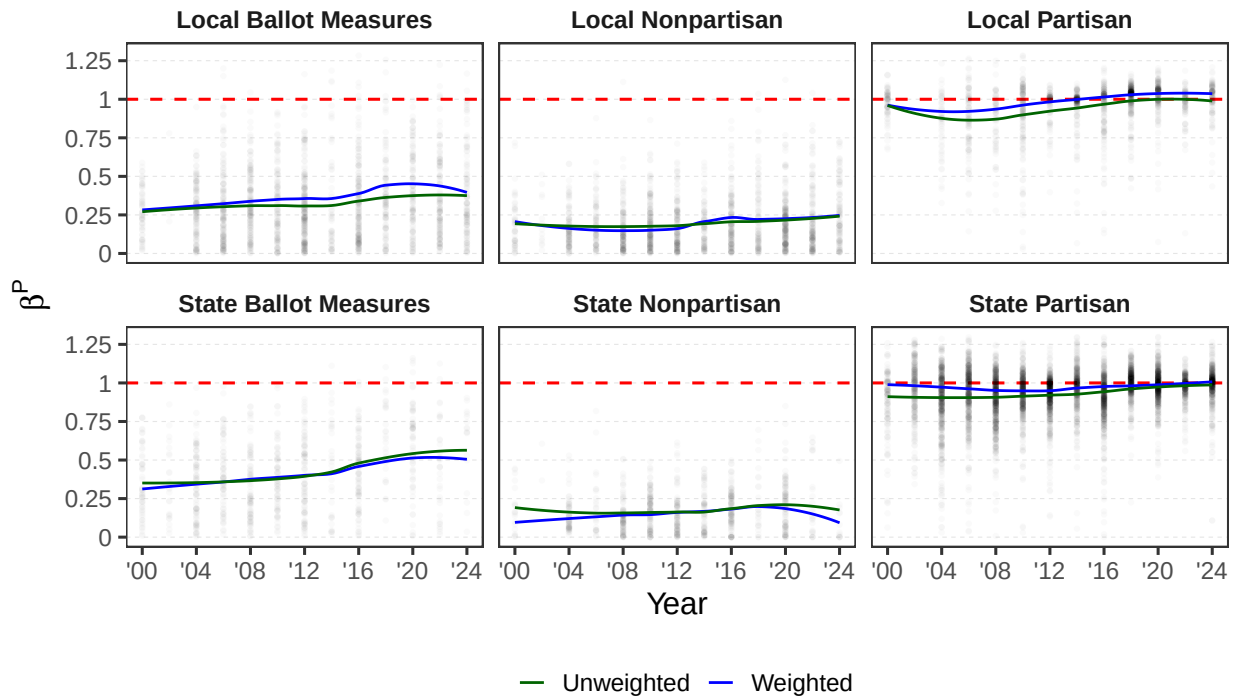


Figure C.2: Nationalization of State and Local Elections Using President and Senate Only As Top-of-the-Ticket Offices

D Main Results by Jurisdiction

In this section, we display our main figure, Figure 2, separately for each state or county that has data on both state and local level contests. Unlike in the main text, where we limit the range of the figure to 2000-2024, here we allow all years to be displayed. We also persist all facets of the plot, despite only some states having data available for each type of contest. Otherwise, the figures are constructed in the exact same way as in the main text.

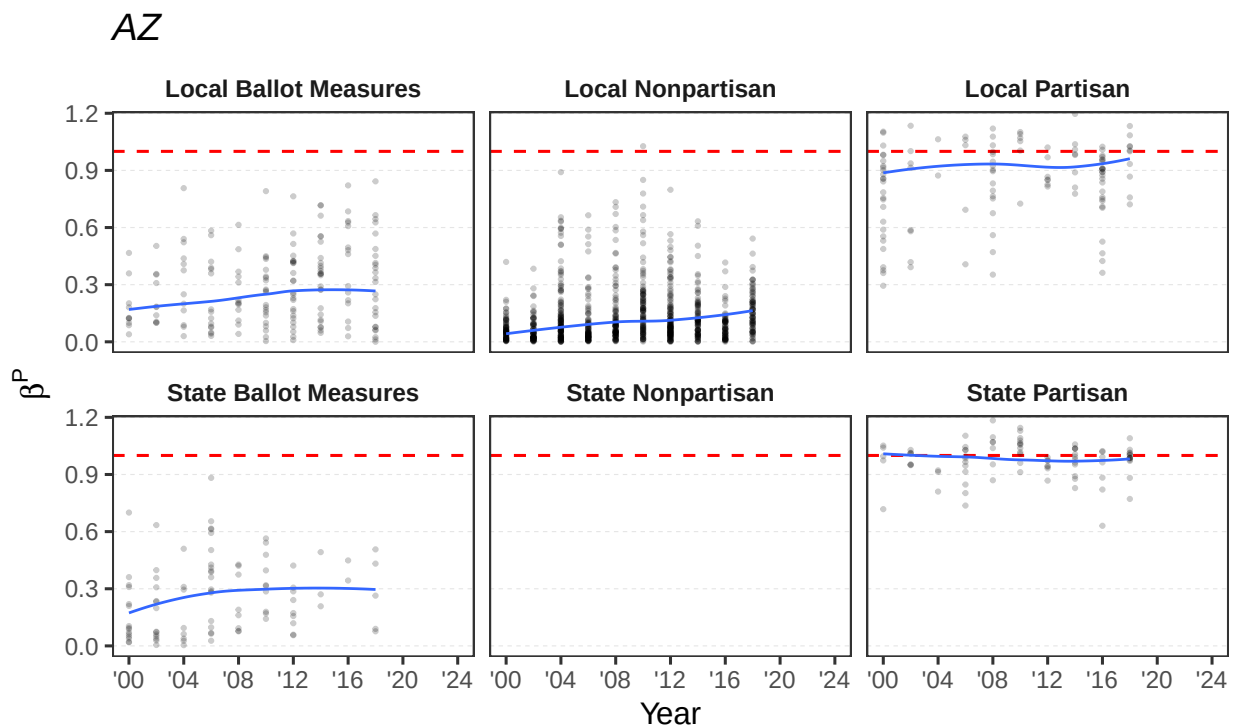


Figure D.1: Main Results: Arizona

MA

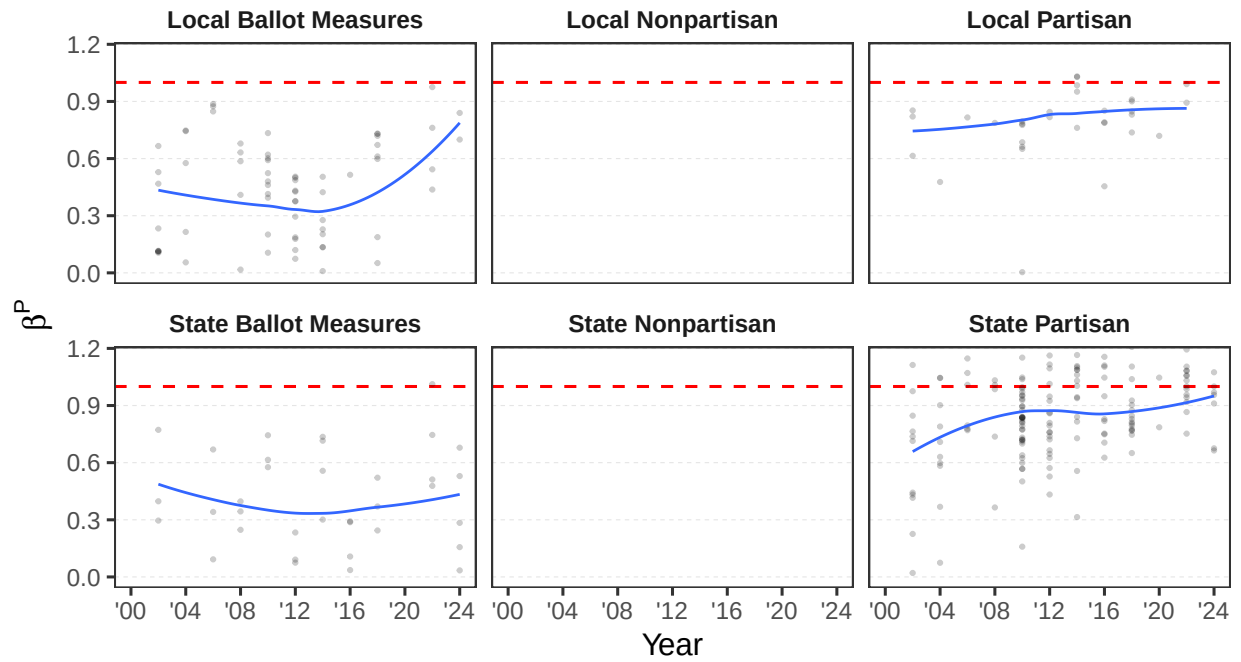


Figure D.2: Main Results: Massachusetts

MD

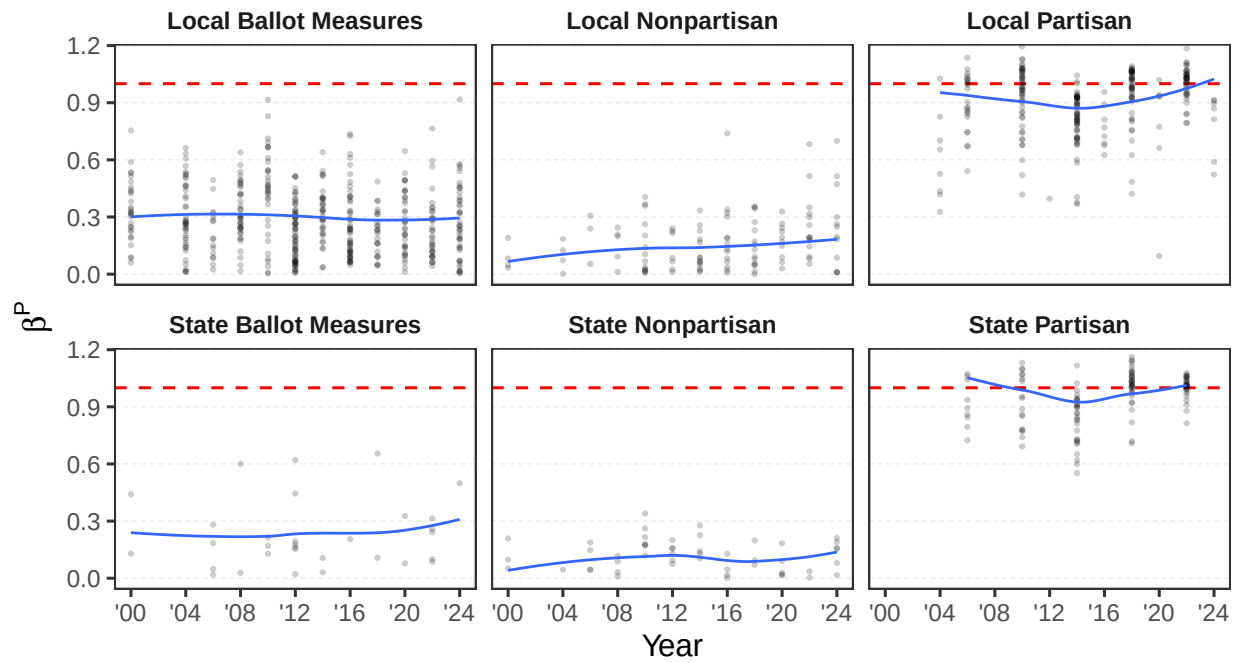


Figure D.3: Main Results: Maryland

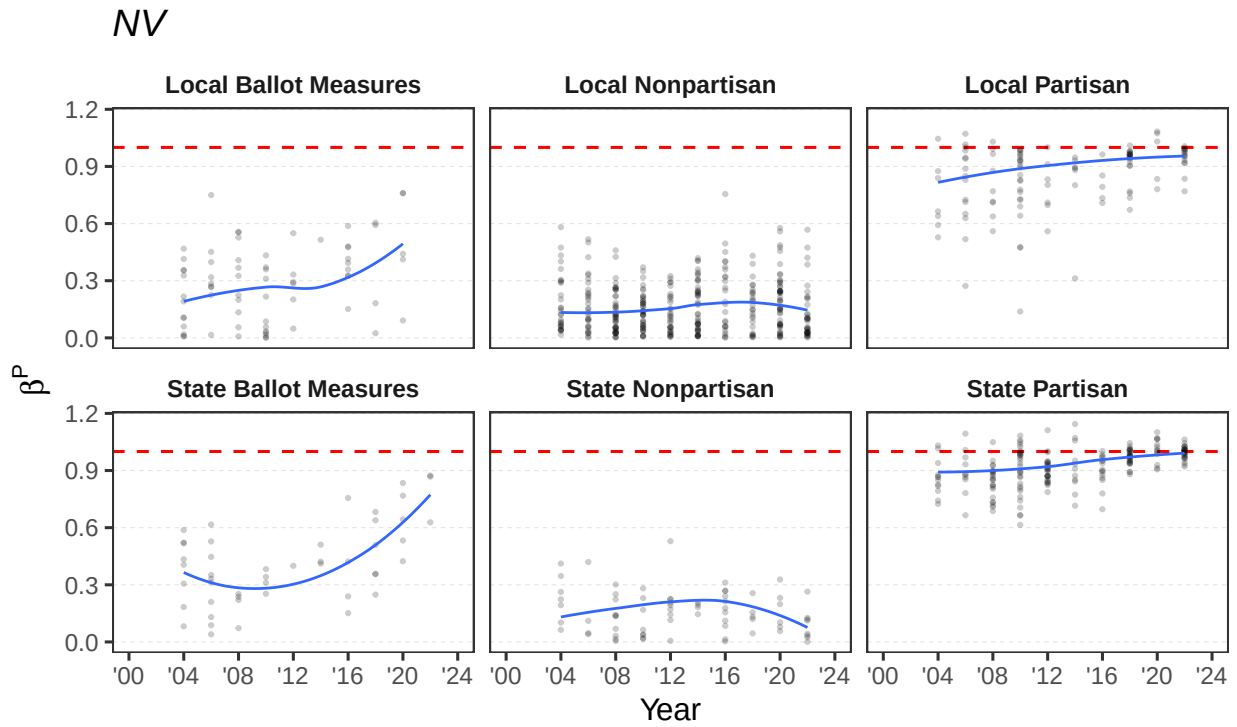


Figure D.4: Main Results: Nevada

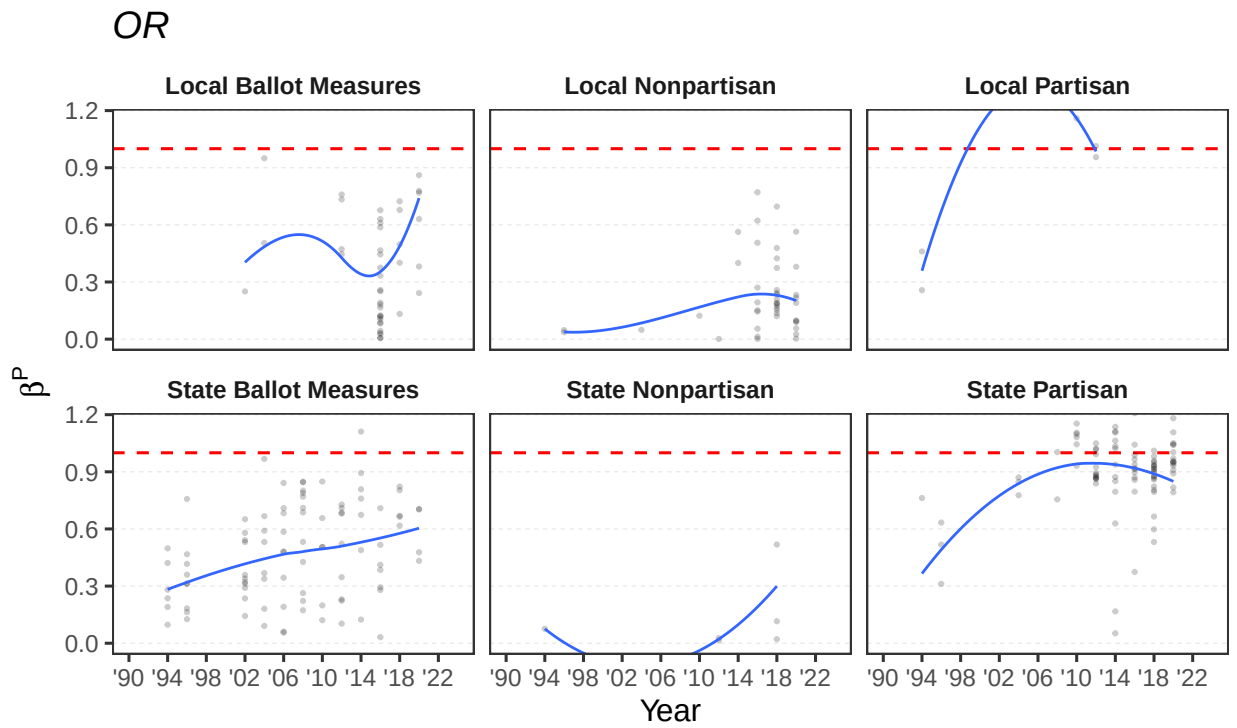


Figure D.5: Main Results: Oregon

RI

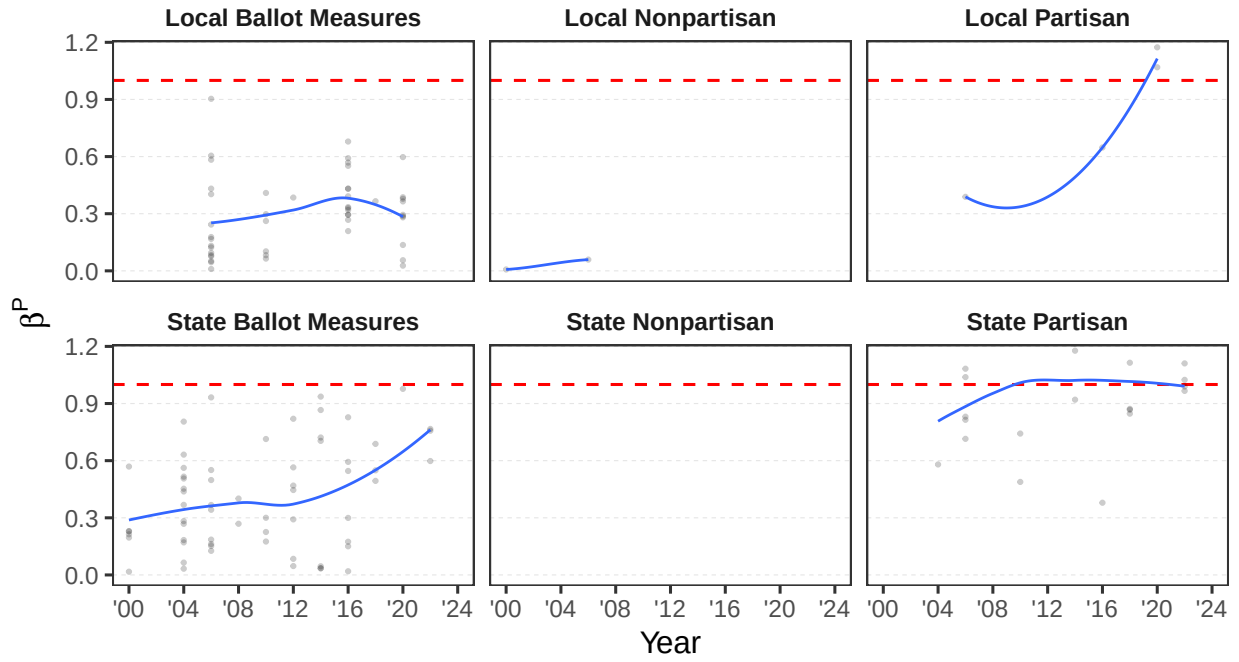


Figure D.6: Main Results: Rhode Island

SC

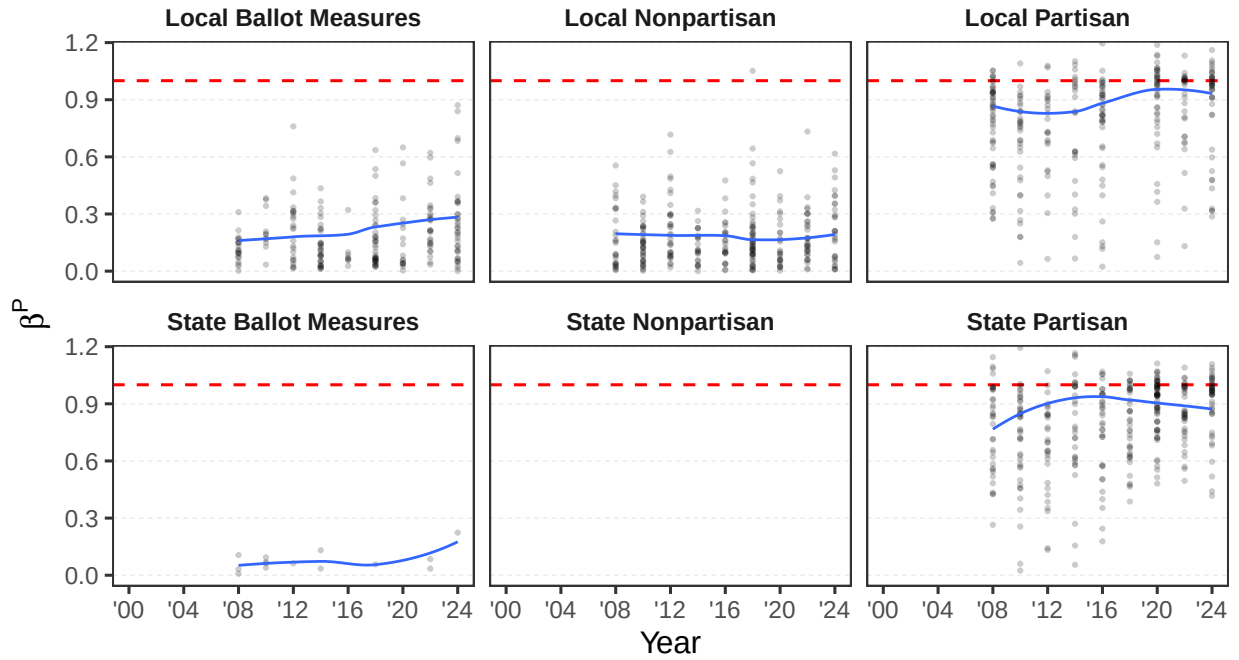


Figure D.7: Main Results: South Carolina

Santa Clara, CA

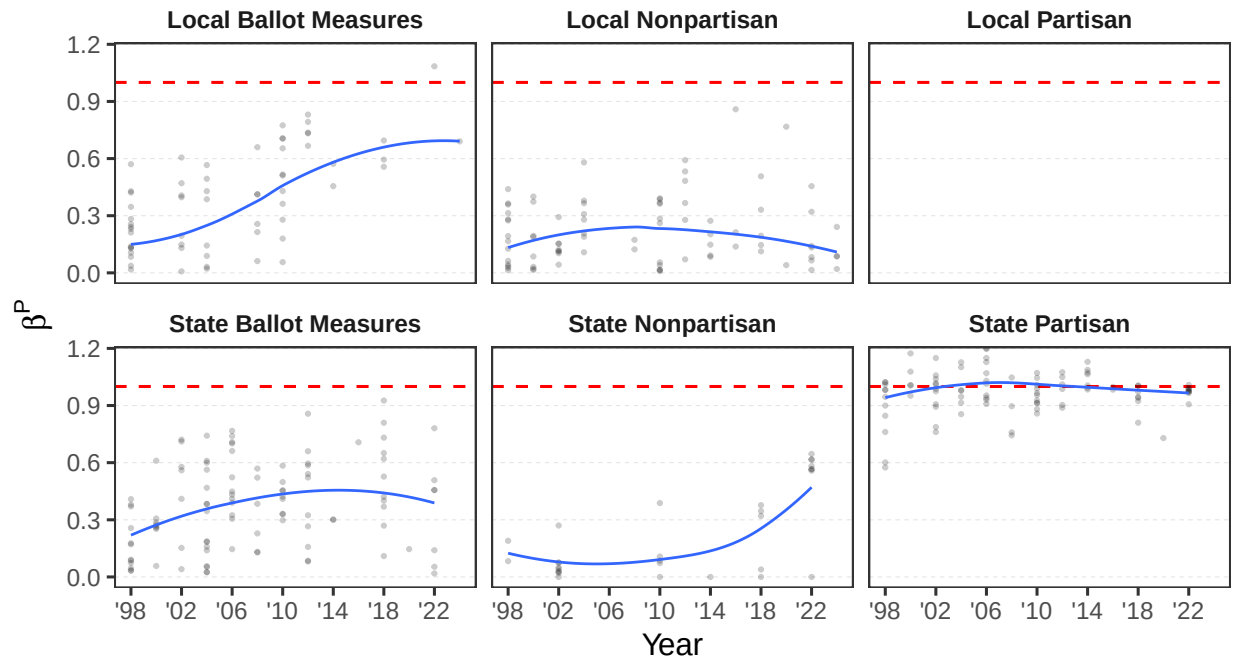


Figure D.8: Main Results: Santa Clara, CA

Marion, IN

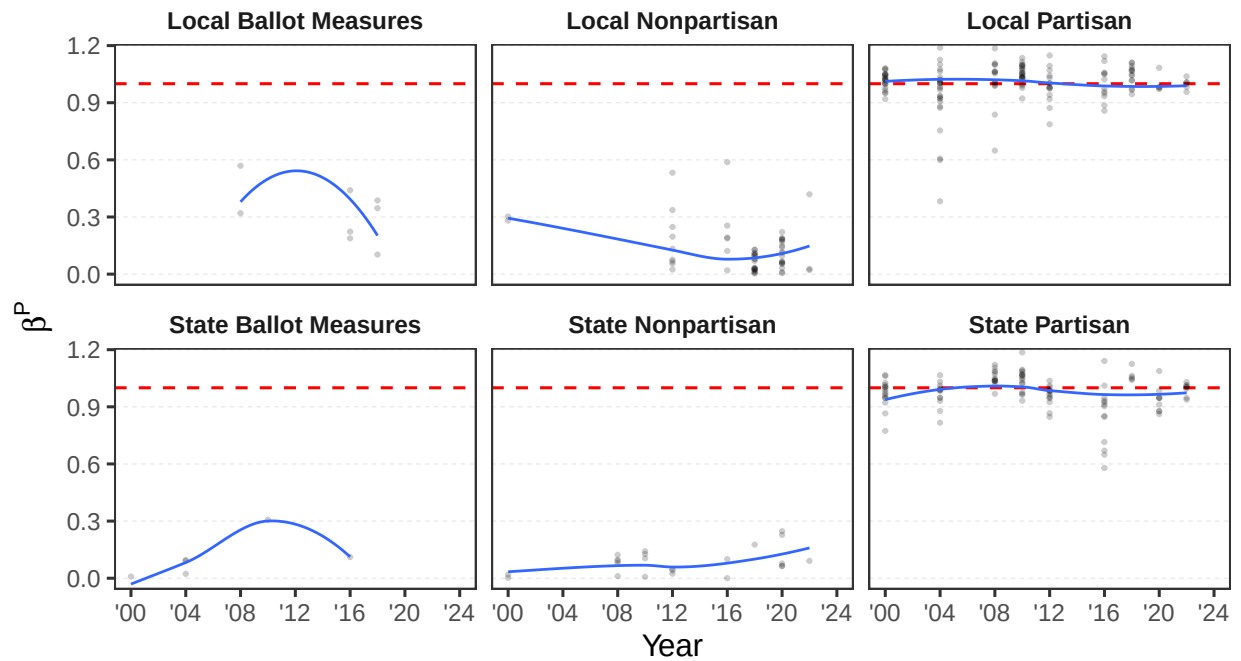


Figure D.9: Main Results: Marion, IN

Cuyahoga, OH

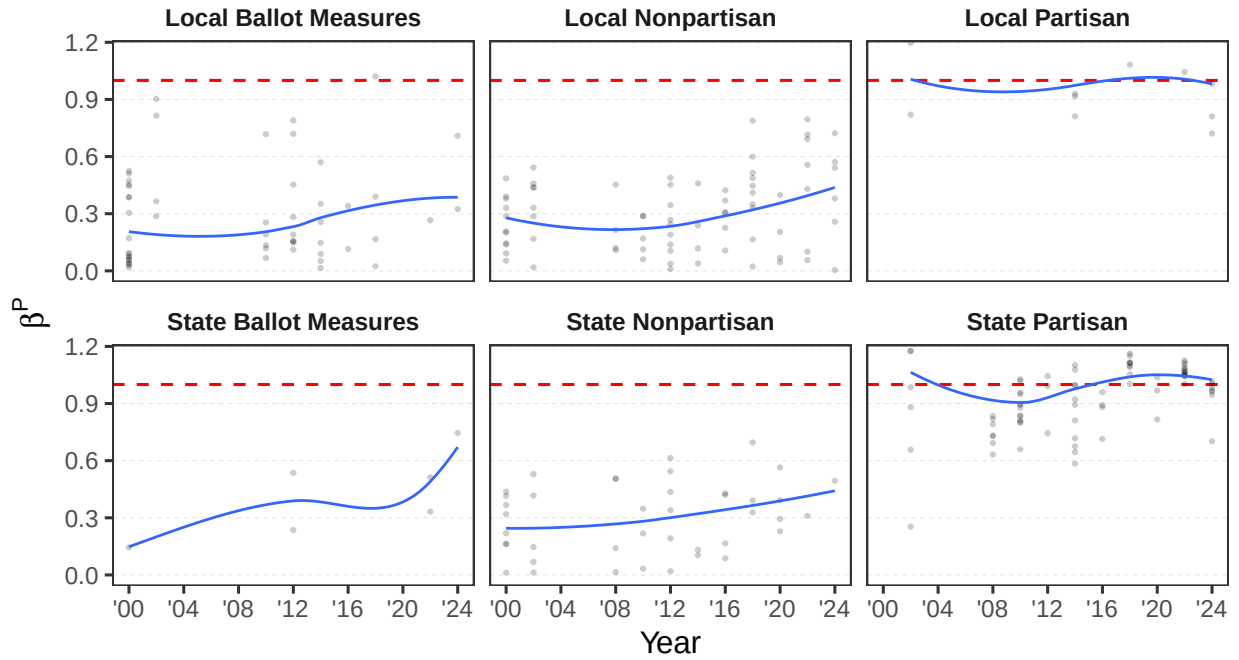


Figure D.10: Main Results: Cuyahoga, OH

Franklin, OH

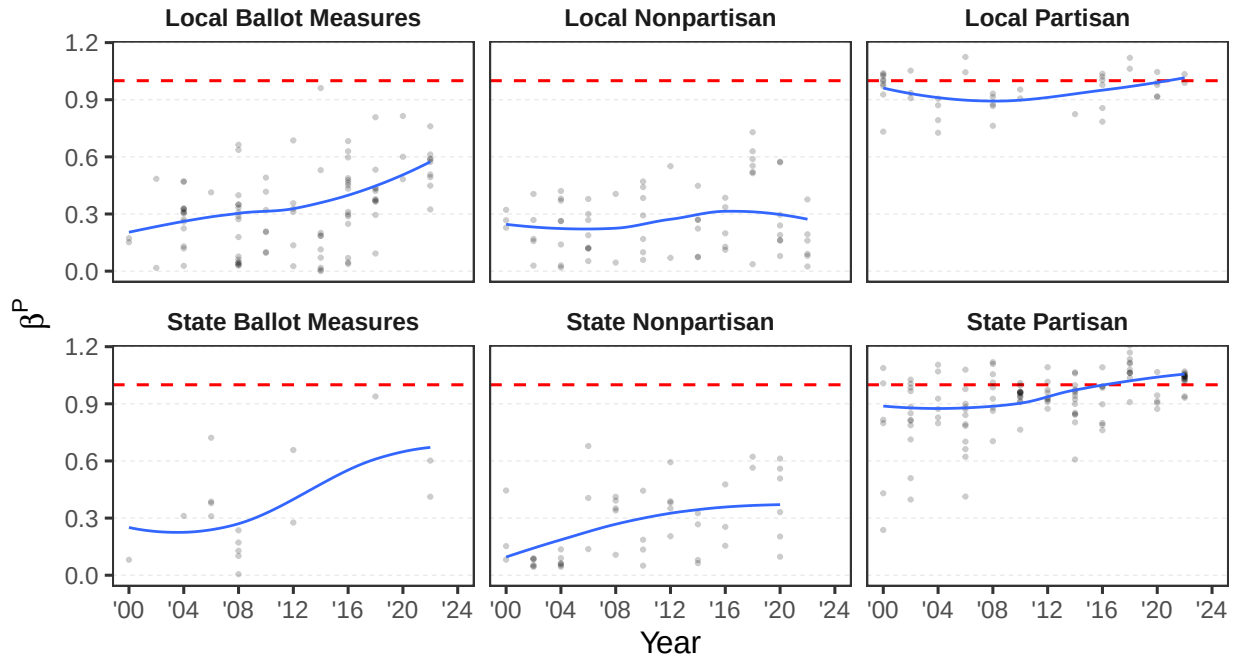


Figure D.11: Main Results: Franklin, OH

Hamilton, OH

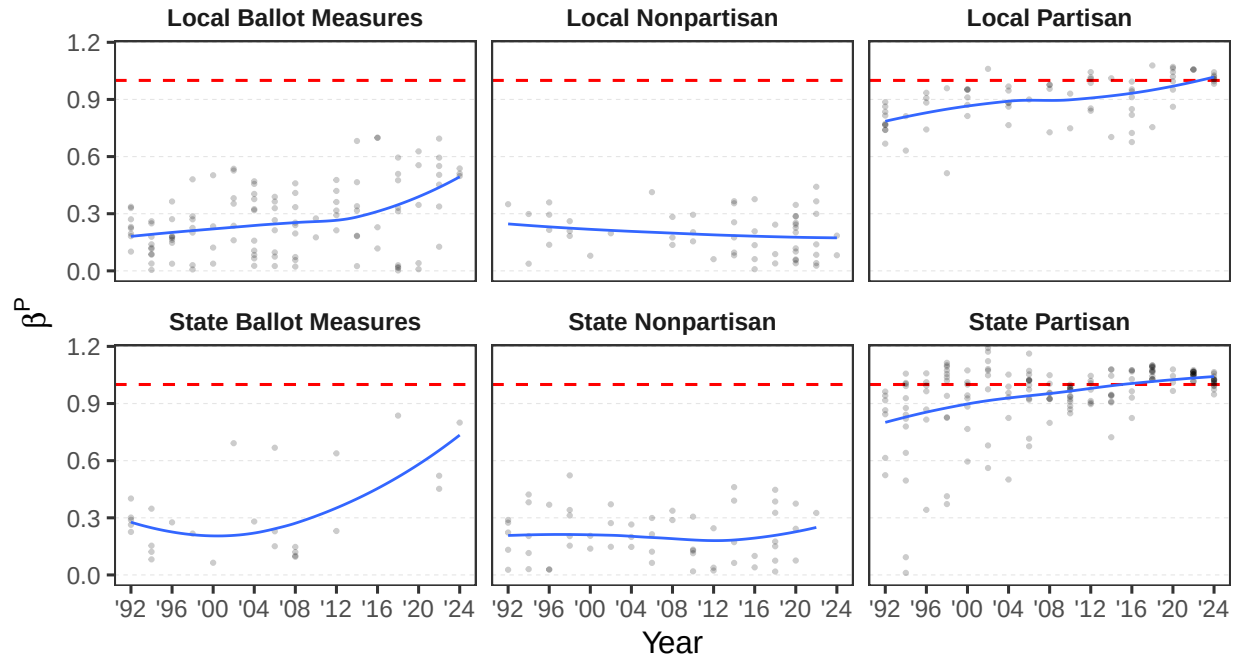


Figure D.12: Main Results: Hamilton, OH

King, WA

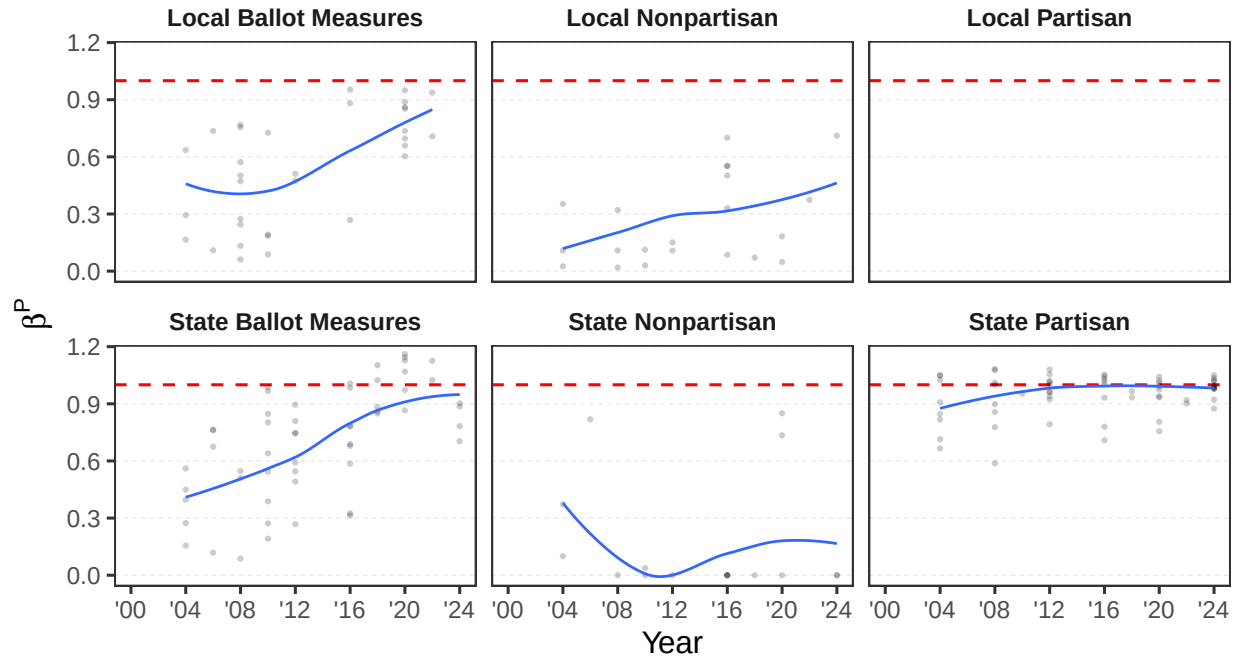


Figure D.13: Main Results: King, WA

Pierce, WA

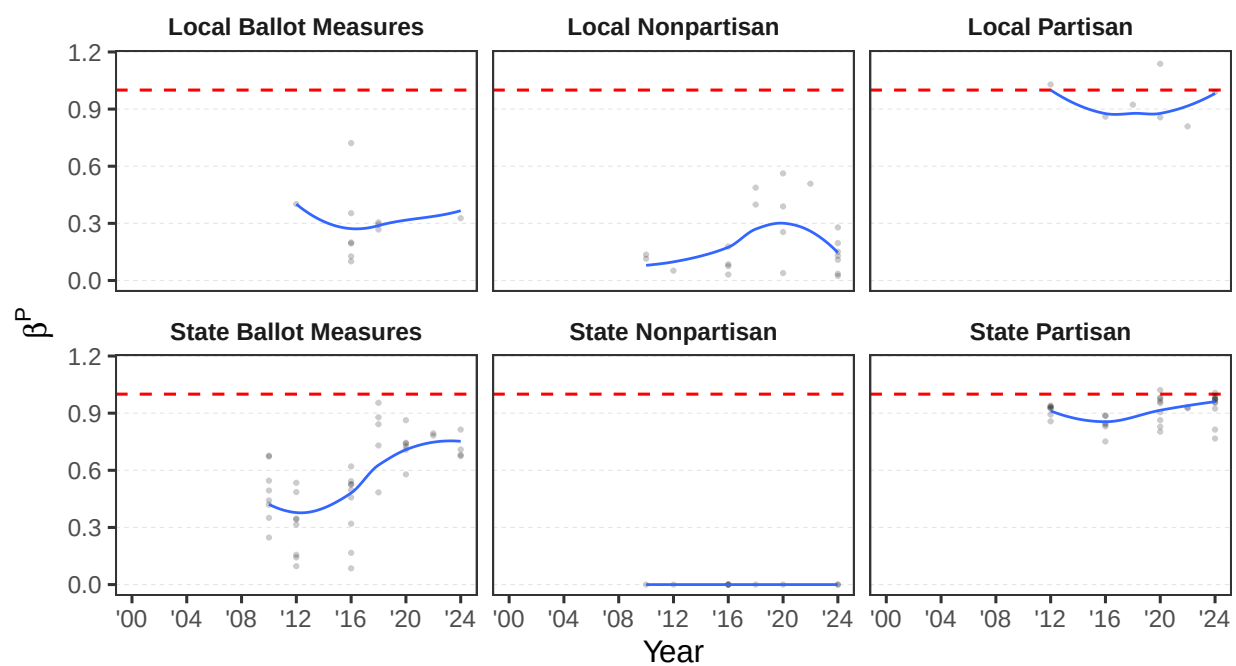


Figure D.14: Main Results: Pierce, WA